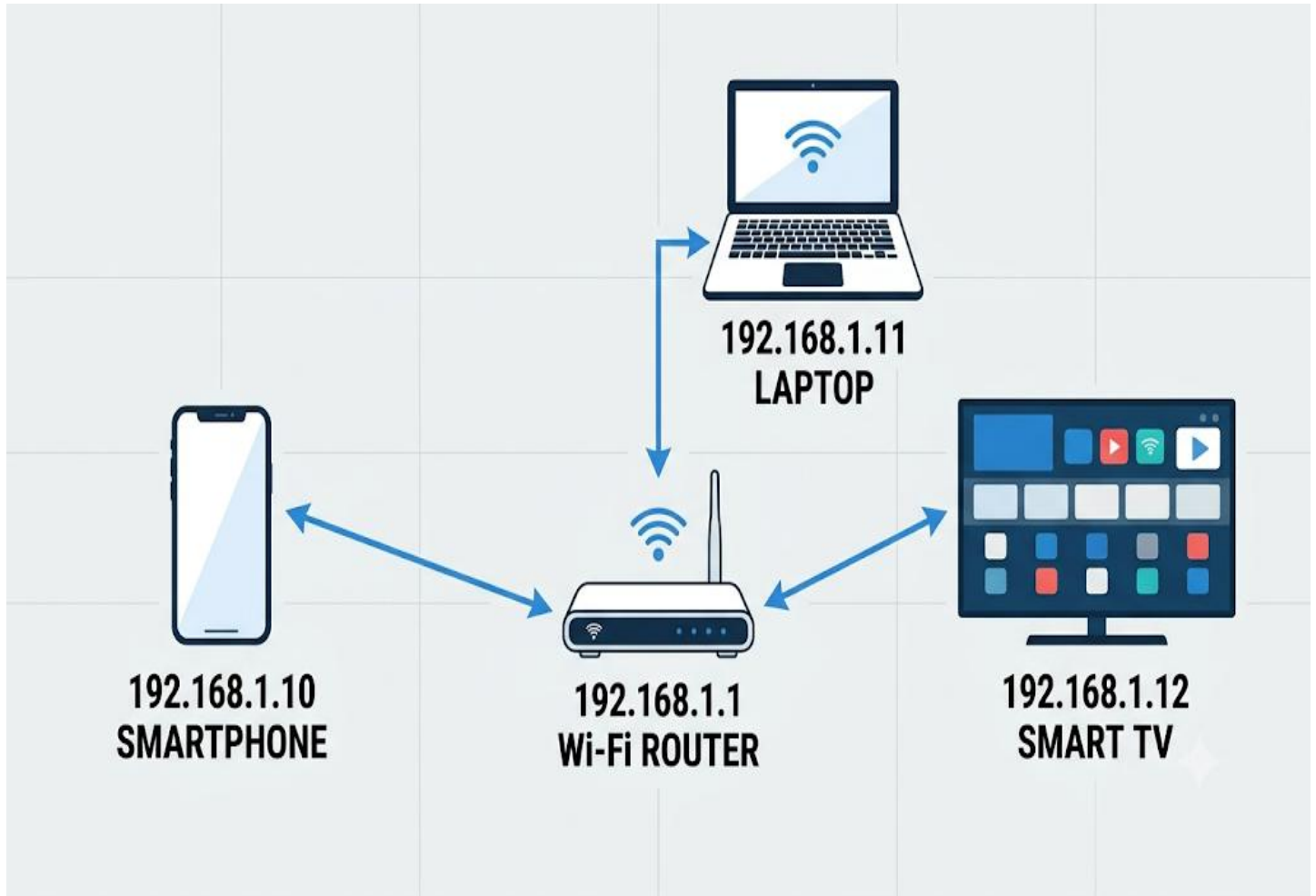


IP ROUTING AND PROTOCOLS

1. IMAGE OF HOME NETWORK SETUP :



2. SENDING A MESSAGE TO A FRIEND IN ANOTHER NETWORK :

WHEN YOU SEND DATA TO SOMEONE OUTSIDE YOUR LOCAL Wi-Fi, IT WORKS LIKE SENDING A PHYSICAL LETTER:

1. **CHECKING THE ADDRESS:** YOUR LAPTOP SEES THAT YOUR FRIEND'S COMPUTER ISN'T IN YOUR LOCAL GROUP. SO, IT SENDS THE PACKET TO YOUR LOCAL ROUTER (ROUTER A).
2. **FIRST HOP (ROUTER A):** ROUTER A LOOKS AT ITS LOOKUP TABLE. IT SEES THAT YOUR FRIEND'S NETWORK IS CONNECTED TO ROUTER B, SO IT PASSES THE MESSAGE TO ROUTER B.
3. **SECOND HOP (ROUTER B):** ROUTER B RECEIVES THE MESSAGE, LOOKS AT ITS LOOKUP TABLE, AND SEES THAT YOUR FRIEND'S LAPTOP IS DIRECTLY PLUGGED INTO IT.
4. **DELIVERED:** ROUTER B HANDS THE MESSAGE STRAIGHT TO YOUR FRIEND'S LAPTOP.

3. FOOD DELIVERY NETWORK ROUTING TABLE :

IMAGINE A CENTRAL COMPUTER DIRECTING TRAFFIC BETWEEN FOOD DELIVERY OFFICES IN THREE CITIES. THE ROUTER USES A SIMPLE CHEAT SHEET TO DIRECT INCOMING TRAFFIC:

TARGET CITY / NETWORK	MASK (SIZE)	SEND TO (NEXT HOP)	EXIT DOOR (INTERFACE)
MUMBAI BRANCH (172.16.10.0)	255.255.255.0	MUMBAI ROUTER (10.100.0.1)	DOOR 1 (ETH0)
DELHI BRANCH (172.16.20.0)	255.255.255.0	DELHI ROUTER (10.100.0.2)	DOOR 2 (ETH1)

TARGET CITY / NETWORK	MASK (SIZE)	SEND TO (NEXT HOP)	EXIT DOOR (INTERFACE)
BENGALURU BRANCH (172.16.30.0)	255.255.255.0	BENGALURU ROUTER (10.100.0.3)	DOOR 3 (ETH2)

4. STATIC VS. DYNAMIC ROUTING (SIMPLE COMPARISON) :

- **STATIC ROUTING (MANUAL SETUP)**
 - *WHAT IT IS:* YOU MANUALLY TYPE OUT EVERY PATH INTO THE ROUTER YOURSELF.
 - *PROS:* FAST, CHEAP, VERY SECURE, DOESN'T WASTE ROUTER MEMORY.
 - *CONS:* IF A CABLE BREAKS, TRAFFIC STOPS COMPLETELY UNTIL A HUMAN FIXES IT MANUALLY.
 - *REAL-WORLD EXAMPLE:* A STANDALONE CASH REGISTER IN A SMALL GROCERY STORE THAT ONLY SENDS DATA TO ONE LOCAL PC.
- **DYNAMIC ROUTING (AUTOMATIC SETUP)**
 - *WHAT IT IS:* ROUTERS CONSTANTLY TALK TO EACH OTHER AND AUTOMATICALLY FIND THE FASTEST, WORKING PATH.
 - *PROS:* IF A CONNECTION DROPS, IT INSTANTLY REROUTES TRAFFIC AROUND THE PROBLEM AUTOMATICALLY.
 - *CONS:* USES MORE MEMORY AND PROCESSING POWER.
 - *REAL-WORLD EXAMPLE:* APPS LIKE **ZOMATO** OR **GOOGLE MAPS**. IF ONE ROAD OR SERVER GOES DOWN, THE SYSTEM INSTANTLY REDIRECTS YOUR ORDER THROUGH A WORKING ROUTE SO YOUR APP DOESN'T CRASH.

5. SIMPLE ROUTER CONFIGURATION BREAKDOWN :

HERE IS WHAT THOSE ROUTER COMMAND LINES ACTUALLY MEAN IN PLAIN ENGLISH:

- ROUTER OSPF 1
 - **TRANSLATION:** *"HEY ROUTER, TURN ON THE DYNAMIC ROUTING FEATURE CALLED OSPF."*
- ROUTER-ID 1.1.1.1
 - **TRANSLATION:** *"GIVE THIS ROUTER THE NAME TAG '1.1.1.1' SO OTHER ROUTERS RECOGNIZE IT."*
- NETWORK 10.0.0.0 0.0.0.3 AREA 0
 - **TRANSLATION:** *"SHARE THE ROAD CONNECTED TO 10.0.0.0 WITH NEIGHBOR ROUTERS."*
- NETWORK 192.168.10.0 0.0.0.255 AREA 0
 - **TRANSLATION:** *"LET NEIGHBORING ROUTERS KNOW ABOUT THE OFFICE DEVICES ON THE 192.168.10.0 NETWORK."*